UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Chapter 9

FUZZY LOGIC

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What is fuzzy thinking?

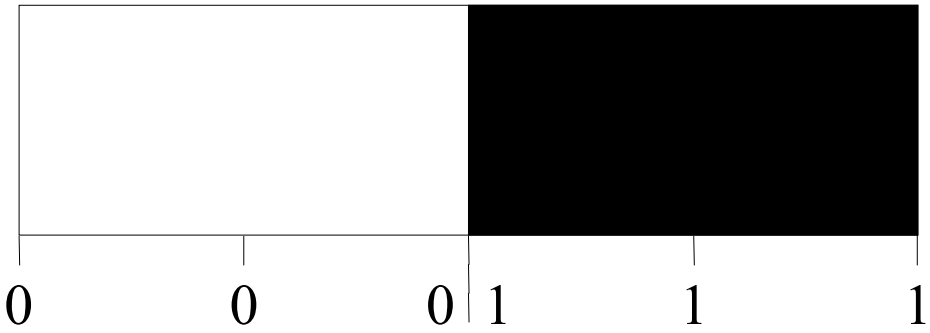
- Experts rely on common sense when they solve problems.
- How can we represent expert knowledge that uses vague and ambiguous terms in a computer?
- Fuzzy logic is not logic that is fuzzy, but logic that is used to describe fuzziness. Fuzzy logic is the theory of fuzzy sets, sets that calibrate vagueness.
- Fuzzy logic is based on the idea that all things admit of degrees. Temperature, height, speed, distance, beauty – all come on a sliding scale. The motor is running really hot. Tom is a very tall guy.

What is fuzzy thinking?

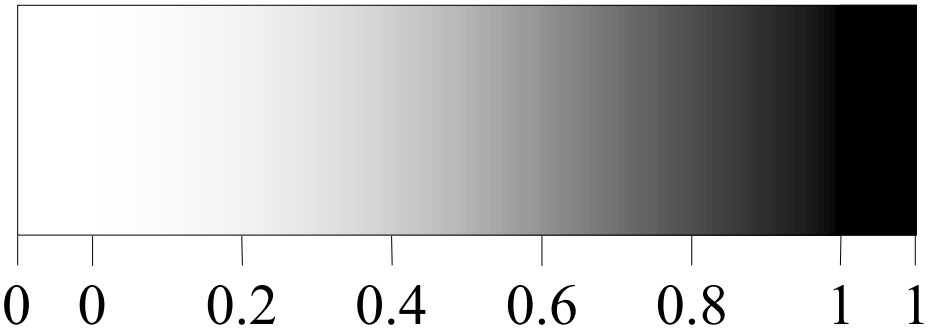
- In 1965 Lotfi Zadeh, published his famous paper “Fuzzy sets”.
- Zadeh extended the work on possibility theory into a formal system of mathematical logic, and introduced a new concept for applying natural language terms.
- This new logic for representing and manipulating fuzzy terms was called fuzzy logic, and Zadeh became the Master/Father of fuzzy logic.

What is Fuzzy Logic?

- Fuzzy logic is a set of mathematical principles for knowledge representation based on degrees of membership rather than on crisp membership of classical binary logic.
- Unlike two-valued Boolean logic, fuzzy logic is multi-valued. It deals with degrees of membership and degrees of truth.



(*a*) Boolean Logic.



(*b*) Multi-valued Logic.

Fuzzy Set?

- The classical example in fuzzy sets is tall men. The elements of the fuzzy set "tall men" are all men, but their degrees of membership depend on their height.

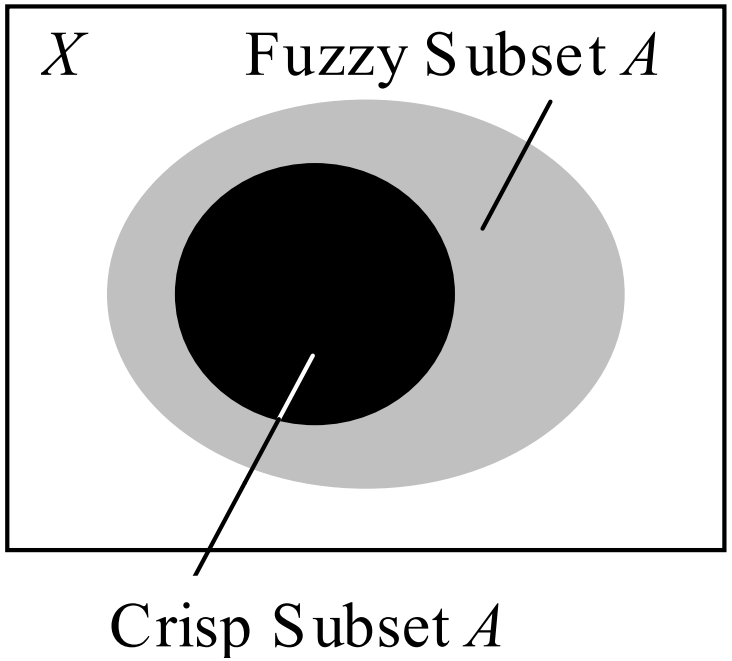
Name	Height, cm	Degree of Membership	
		Crisp	*Fuzzy*
Chris	208	1	1.00
Mark	205	1	1.00
John	198	1	0.98
Tom	181	1	0.82
David	179	0	0.78
Mike	172	0	0.24
Bob	167	0	0.15
Steven	158	0	0.06
Bill	155	0	0.01
Peter	152	0	0.00

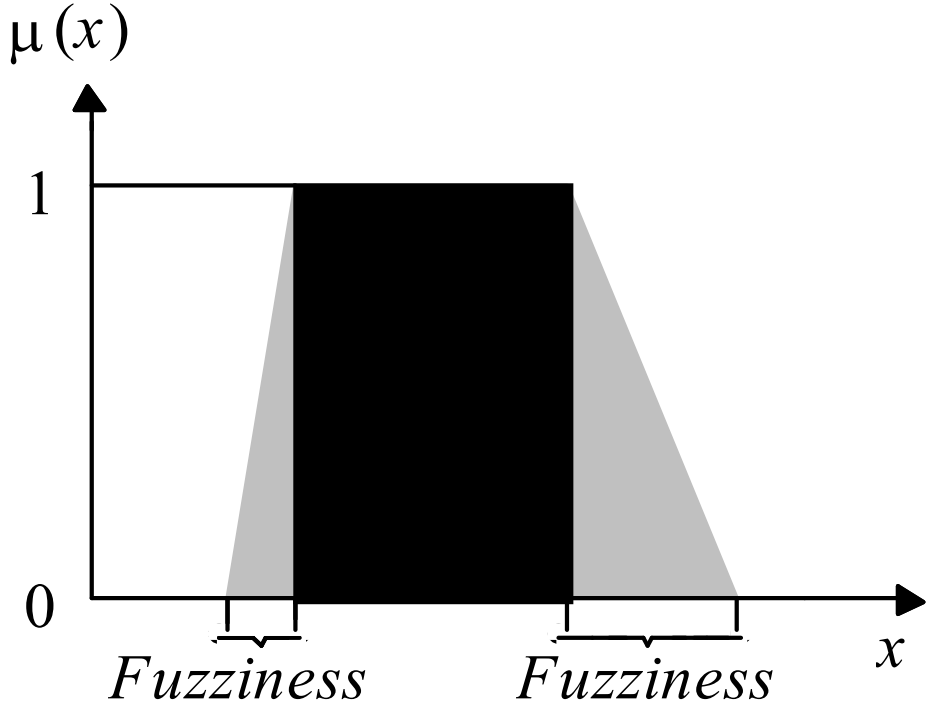
Crisp set asks the question: Is the man tall? : Tall men are above 180, and not tall men are below 180.

Fuzzy set asks the question: How tall is the man?: The tall is partial membership in the fuzzy set, Tom is 0.82 tall.

Crisp and fuzzy subsets

Typical functions that can be used to represent a fuzzy set are sigmoid, gaussian and pi. However, these functions increase the time of computation. Therefore, in practice, most applications use linear fit functions.





Fuzzy rules

Represent the stopping distance rules in a fuzzy form:

Rule: 1
IF speed is fast
THEN stopping_distance is long

Rule: 2
IF speed is slow
THEN stopping_distance is short

A fuzzy rule can have multiple antecedents, for example:

IF project_duration is long
AND project_staffing is large
AND project_funding is inadequate
THEN risk is high

The multiple consequent of a fuzzy rule:

IF temperature is hot
THEN hot_water is reduced;
cold_water is increased

Fuzzy Inference System

- Fuzzy inference is the process of formulating the mapping from a given input to an output using fuzzy theory of fuzzy sets.
- The process of fuzzy inference:
  - membership functions,
  - fuzzy logic operators, and
  - fuzzy rules.
- Two known types of fuzzy inference systems
  - Mamdani-type
  - Sugeno-type.

Mamdani's Fuzzy Inference

- Mamdani's fuzzy inference method is the most commonly seen fuzzy methodology and was among the first control systems built using fuzzy set theory.
- It was proposed in 1975 by Ebrahim Mamdani [Mam75] as an attempt to control a steam engine and boiler combination by synthesizing a set of linguistic control rules obtained from experienced human operators.
- Mamdani's effort was based on Lotfi Zadeh's 1973 paper on fuzzy algorithms for complex systems and decision processes [Zad73].

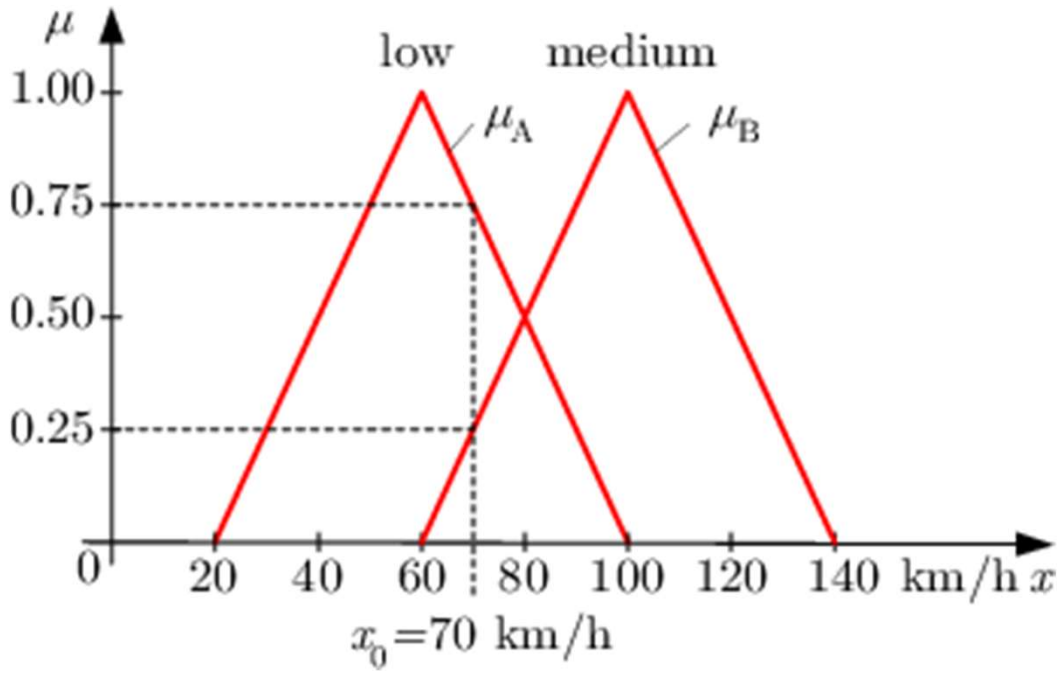
Mamdani's Fuzzy Inference

Mamdani-type inference process is performed in four steps:

1. Fuzzification
2. Rule evaluation
3. Aggregation
4. Defuzzification

Mamdani - Fuzzification

The first step is to take the and determine the degree (membership degree) to which these inputs belong to each of the appropriate fuzzy sets.



Mamdani – Rule evaluation

If the antecedent of a given rule has more than one part, the fuzzy operator is applied to obtain one number that represents the result of the antecedent for that rule.

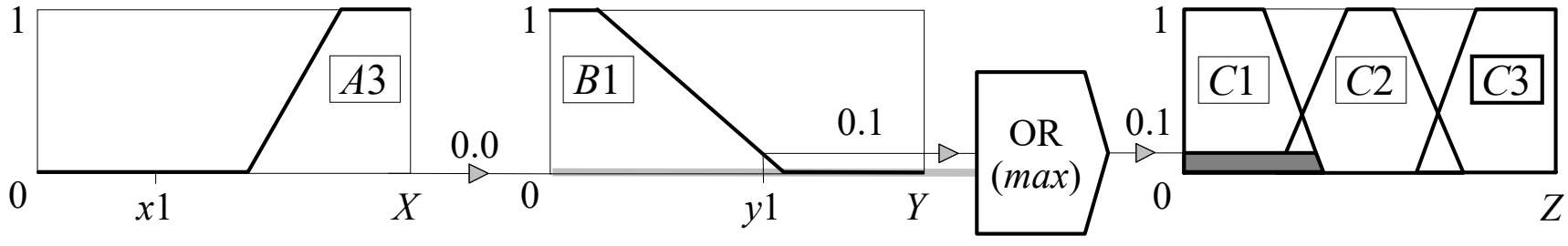
To evaluate the disjunction of the rule antecedents, we use the **OR** fuzzy operation. Typically, fuzzy expert systems make use of the classical fuzzy operation union:

$$\mu A \cup B(x) = \textbf{max} [\mu A(x), \mu B(x)]$$

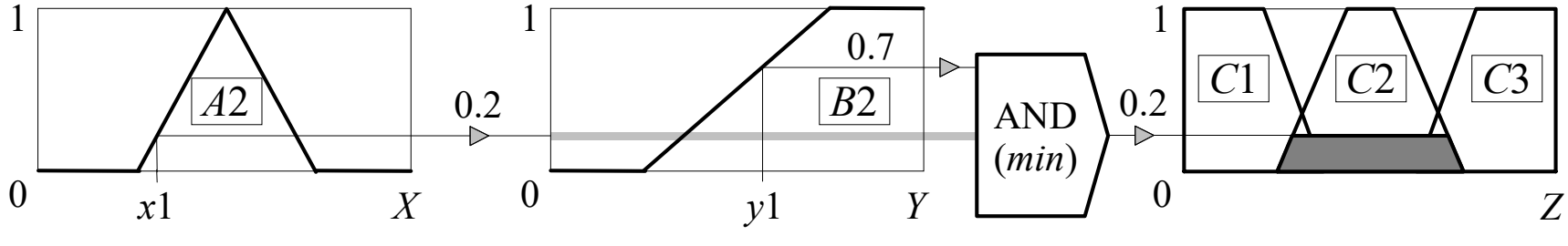
Similarly, in order to evaluate the conjunction of the rule antecedents, we apply the **AND** fuzzy operation intersection:

$$\mu A \cap B(x) = \textbf{min} [\mu A(x), \mu B(x)]$$

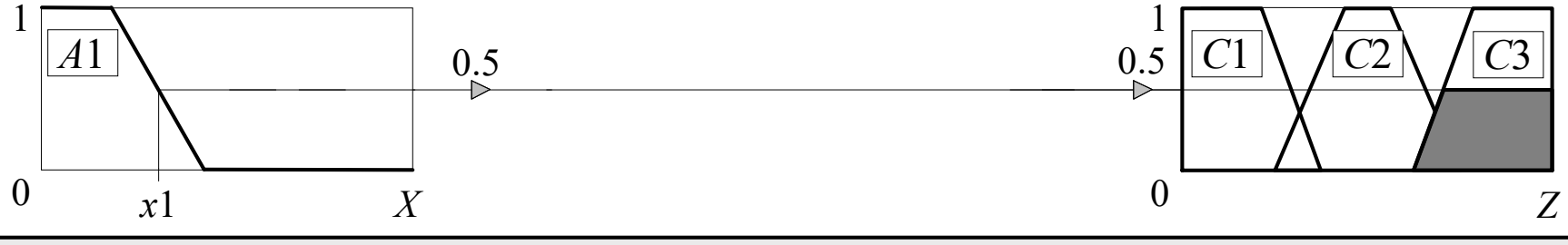
Mamdani – Rule evaluation



Rule 1: IF x is $A3$ (0.0) OR y is $B1$ (0.1) THEN z is $C1$ (0.1)



Rule 2: IF x is $A2$ (0.2) AND y is $B2$ (0.7) THEN z is $C2$ (0.2)

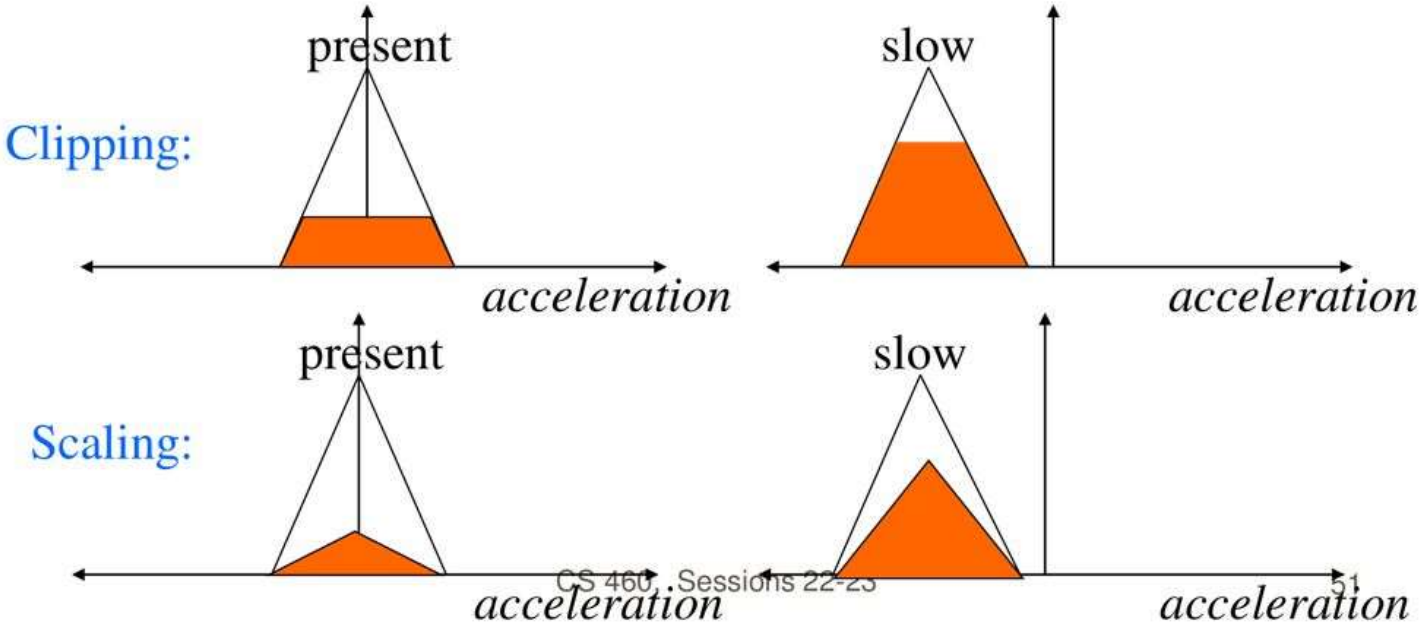


Rule 3: IF x is $A1$ (0.5) THEN z is $C3$ (0.5)

Mamdani – Rule evaluation

Scaling vs. Clipping

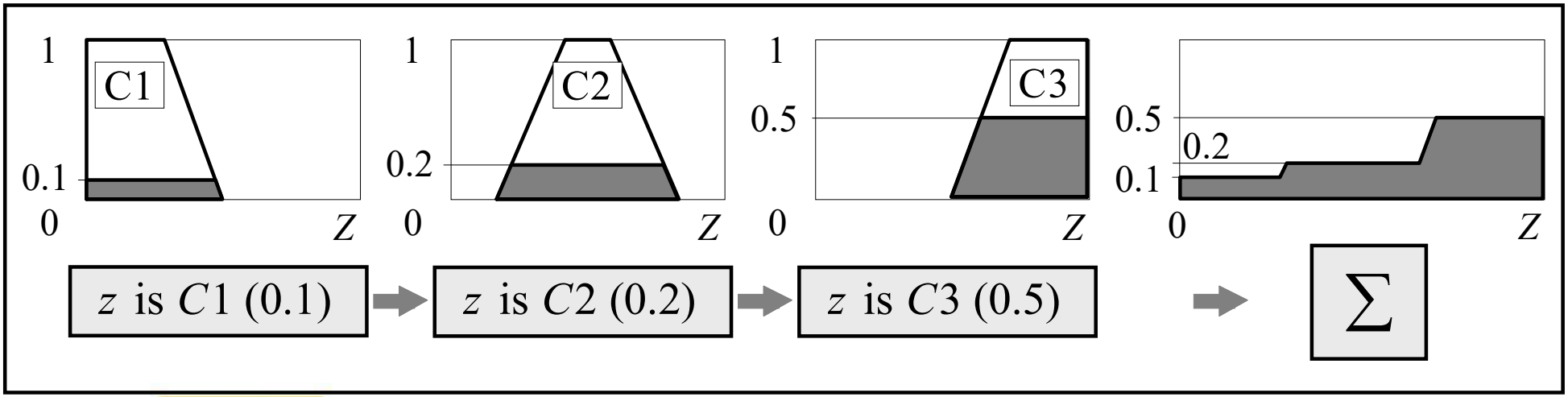
Instead of clipping, another approach is to scale the fuzzy set
By the belief in the premises



Mamdani – Aggregation

Aggregation is the process of unification of the outputs of all rules. We take the membership functions of all rule consequents previously clipped or scaled and combine them into a single fuzzy set.

The input of the aggregation process is the list of clipped or scaled consequent membership functions, and the output is one fuzzy set for each output variable.



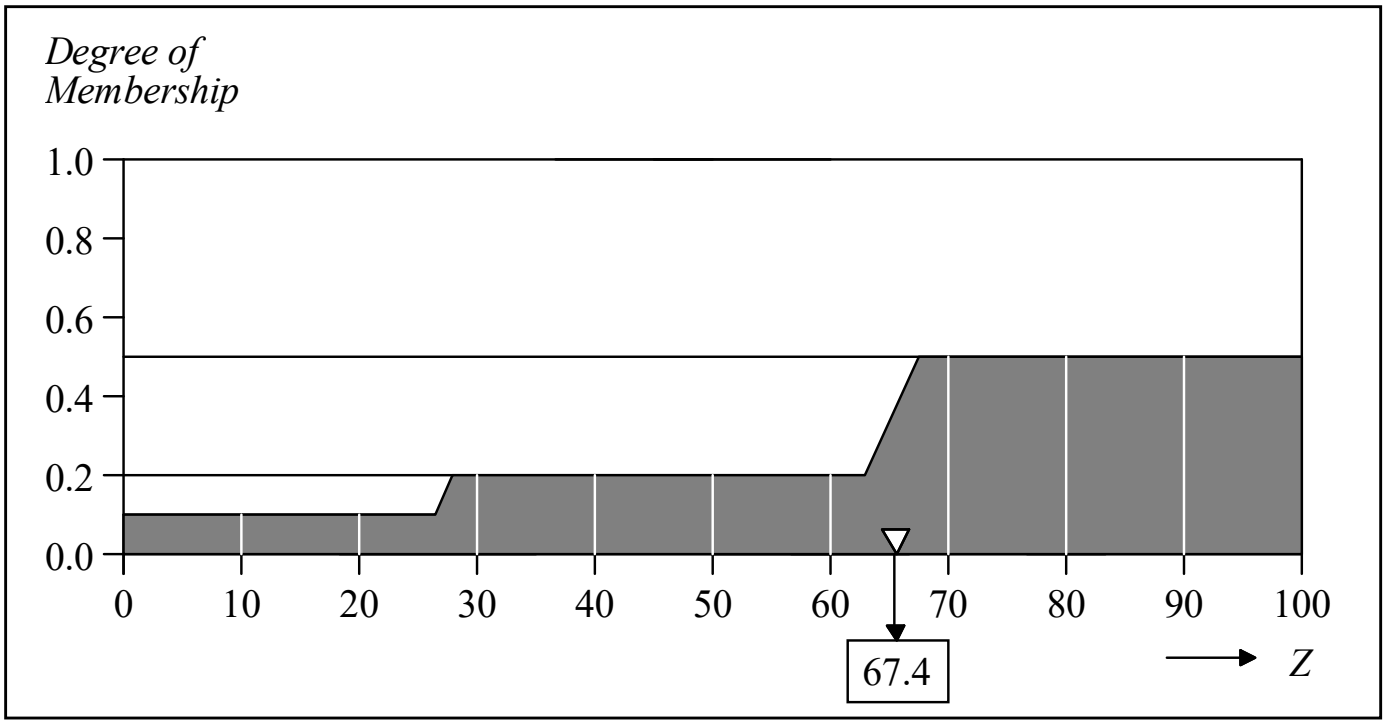
Mamdani – Defuzzification

Defuzzification process is the aggregate output fuzzy set and the output is a single number.

There are several defuzzification methods, but probably the most popular one is the centroid technique. It finds the point where a vertical line would slice the aggregate set into two equal masses. Mathematically this **Centre of Gravity (COG)** can be expressed as:

$$COG = \frac{\int_a^b \mu_A(x)\, x\, dx}{\int_a^b \mu_A(x)\, dx}$$

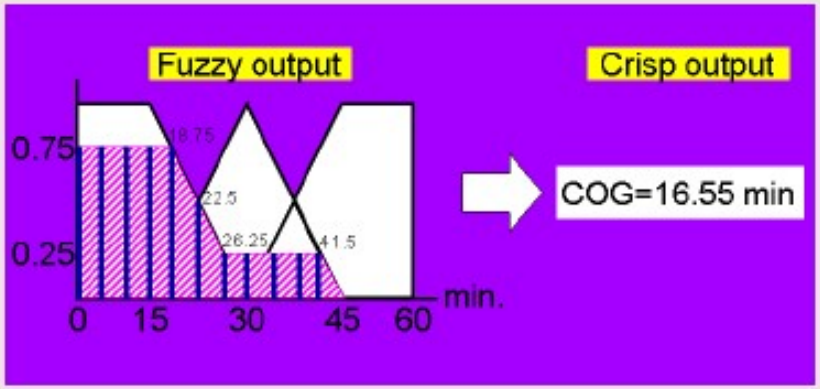
Mamdani – Defuzzification



$$COG = \frac{(0+10+20)\times 0.1+(30+40+50+60)\times 0.2+(70+80+90+100)\times 0.5}{0.1+0.1+0.1+0.2+0.2+0.2+0.2+0.5+0.5+0.5+0.5} = 67.4$$

Mamdani – Defuzzification

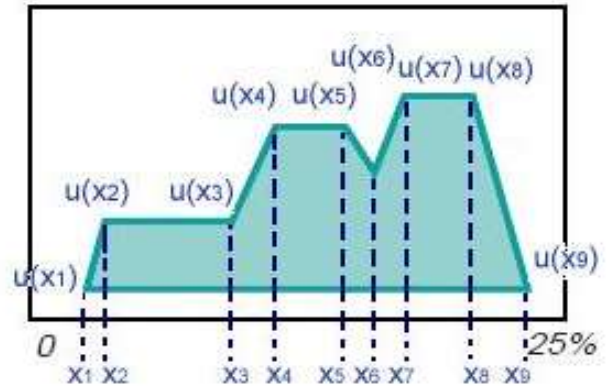
Precision COG



$$COG = \frac{(0.75 \cdot 0) + (0.75 \cdot 5) + (0.75 \cdot 10) + (0.75 \cdot 15) + (0.75 \cdot 18.75) + (0.5 \cdot 22.5) + (0.25 \cdot 26.25) + (0.25 \cdot 30) + (0.25 \cdot 35) + (0.25 \cdot 40) + (0.25 \cdot 41.5) + (0 \cdot 45)}{0.75 + 0.75 + 0.75 + 0.75 + 0.75 + 0.5 + 0.25 + 0.25 + 0.25 + 0.25 + 0.25 + 0} = 16.55$$

Approximate COG

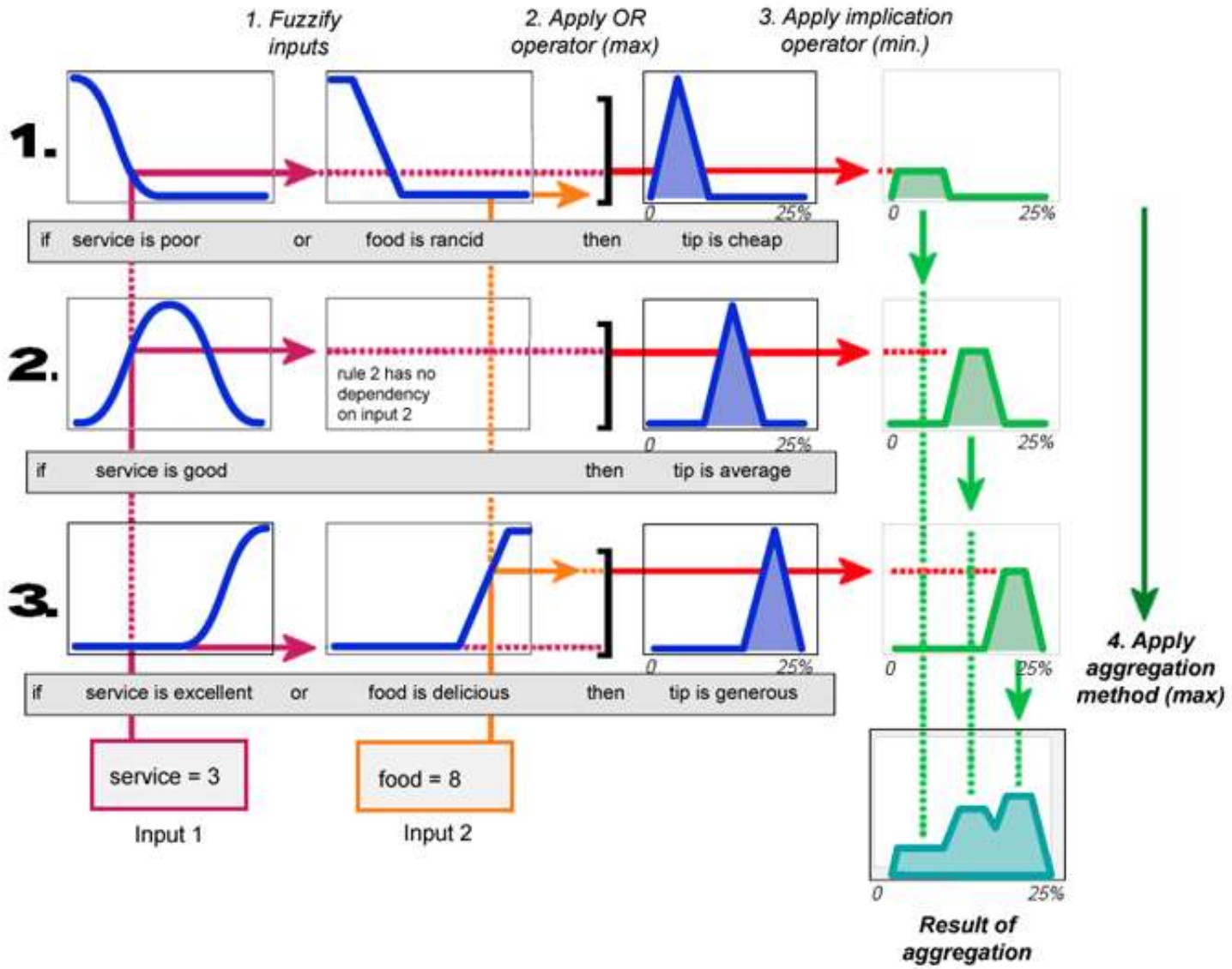
5. Defuzzify the aggregate output (centroid)



$$g = \frac{\sum_{i=1}^{9} x_i \cdot u(x_i)}{\sum_{i=1}^{9} u(x_i)} = 16,7$$

tip= 16,7%

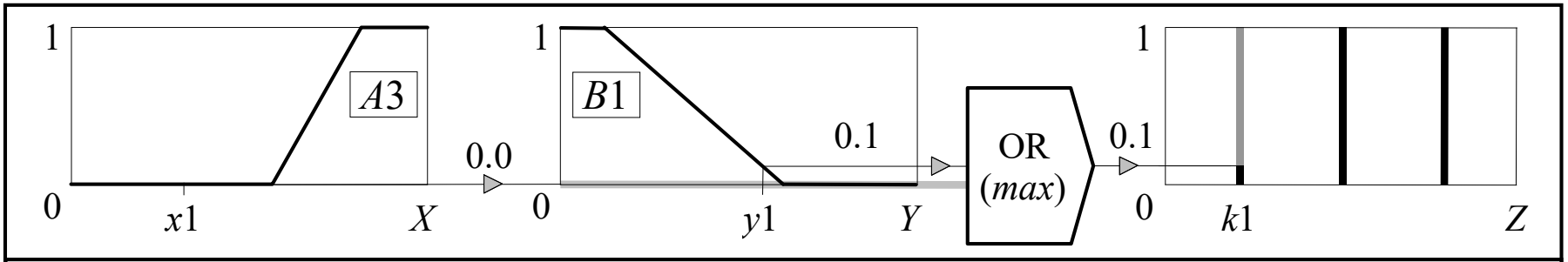
Mamdani – Overview



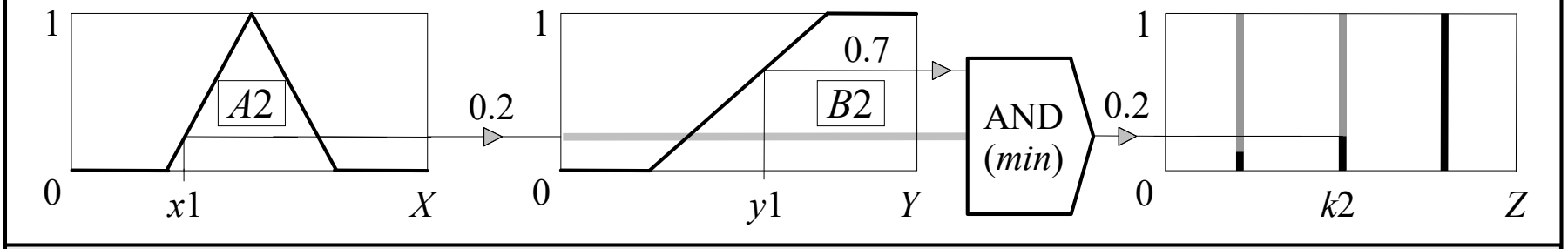
Sugeno Fuzzy Inference

- Mamdani-style inference, as we have just seen, requires us to find the centroid of a two-dimensional shape by integrating across a continuously varying function. In general, this process is not computationally efficient.
- Michio Sugeno suggested to use a single spike, a singleton, as the membership function of the rule consequent. A singleton, or more precisely a fuzzy singleton, is a fuzzy set with a membership function that is unity at a single particular point on the universe of discourse and zero everywhere else.

Sugeno - Rule evaluation



Rule 1: IF *x* is *A*3 (0.0) OR *y* is *B*1 (0.1) THEN *z* is *k*1 (0.1)

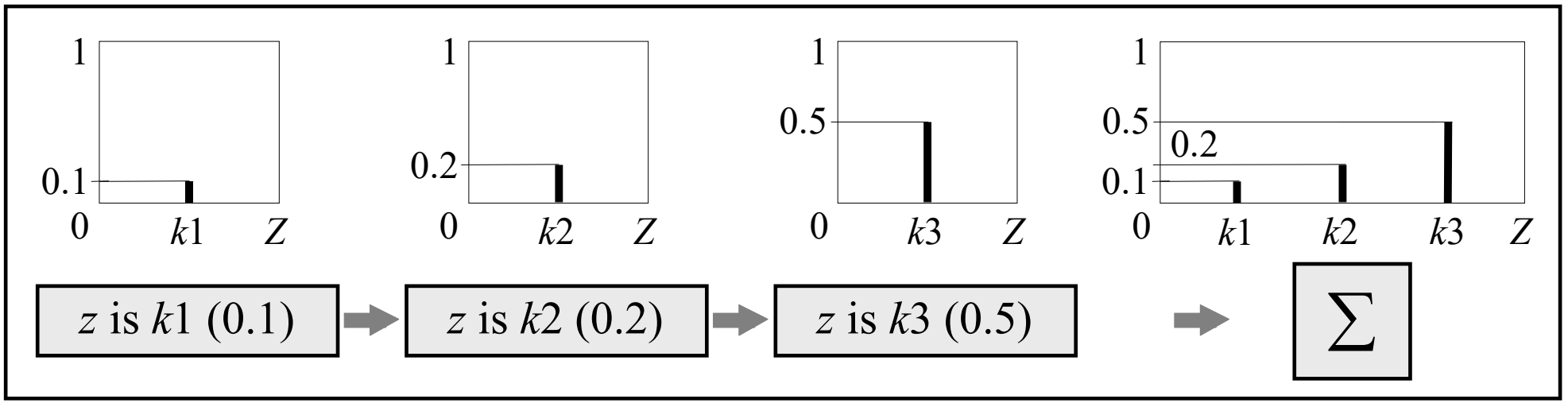


Rule 2: IF *x* is *A*2 (0.2) AND *y* is *B*2 (0.7) THEN *z* is *k*2 (0.2)

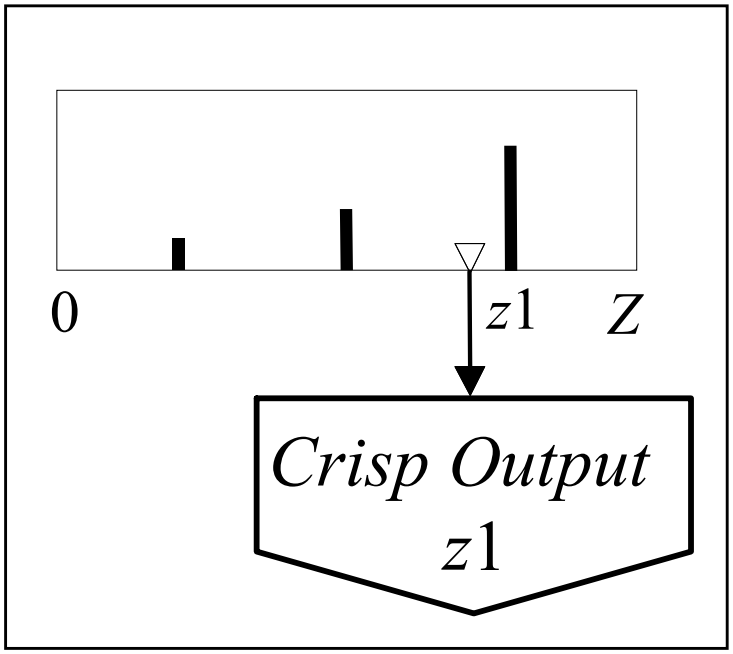


Rule 3: IF *x* is *A*1 (0.5) THEN *z* is *k*3 (0.5)

Sugeno – Aggregation



Sugeno – Defuzzification



$$WA = \frac{\mu(k1)\times k1 + \mu(k2)\times k2 + \mu(k3)\times k3}{\mu(k1) + \mu(k2) + \mu(k3)} = \frac{0.1\times 20 + 0.2\times 50 + 0.5\times 80}{0.1 + 0.2 + 0.5} = 65$$

Case Study – Autonomous Car Speed

Input:

Rain in mm (A) - Light (A1), Heavy (A2) : 0mm - 5mm
Road in m3 (B) - Dry (B1), Wet (B2) : 0m3 - 5m3

Output:

Speed in kmh(C) - Low (C1), Avg(C2), High(C3) : 0-100kmh

Rules:

IF Rain is Light (A1)	AND	Road is Dry (B1)	THEN Speed is High (C3)
IF Rain is Heavy (A2)	AND	Road is Wet (B2)	THEN Speed is Low (C1)
IF Rain is Heavy (A2)	OR	Road is Wet (B2)	THEN Speed is Avg (C2)

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Case Study – Autonomous Car Speed

Fuzzy Membership Functions:

$$\mu_{A1}(X) = \begin{cases} 1 & 0 \le X \le 1 \\ \dfrac{4-3}{3} & 1 < X < 4 \\ 0 & X \ge 4 \end{cases} \qquad \mu_{A2}(X) = \begin{cases} 0 & X \le 1 \\ \dfrac{X-1}{2} & 1 < X < 3 \\ 1 & X \ge 3 \end{cases}$$

$$\mu_{B1}(Y) = \begin{cases} \dfrac{4-Y}{4} & 0 \le Y < 4 \\ 0 & Y \ge 4 \end{cases} \qquad \mu_{B2}(Y) = \begin{cases} 0 & Y \le 1 \\ \dfrac{Y-1}{3} & 1 < Y < 4 \\ 1 & Y \ge 4 \end{cases}$$

$$\mu_{C1}(Z) = \begin{cases} 0 & Z \le 0 \\ \dfrac{Z-0}{25} & 0 < Z \le 25 \\ \dfrac{50-Z}{25} & 25 < Z < 50 \\ 0 & Z \ge 50 \end{cases} \qquad \mu_{C2}(Z) = \begin{cases} 0 & Z \le 25 \\ \dfrac{Z-25}{25} & 25 < Z \le 50 \\ \dfrac{75-Z}{25} & 50 < Z < 75 \\ 0 & Z \ge 75 \end{cases}$$

$$\mu_{C3}(Z) = \begin{cases} 0 & Z \le 50 \\ \dfrac{Z-50}{25} & 50 < Z \le 75 \\ \dfrac{100-Z}{25} & 75 < Z < 100 \\ 0 & Z \ge 100 \end{cases}$$

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Case Study – Autonomous Car Speed

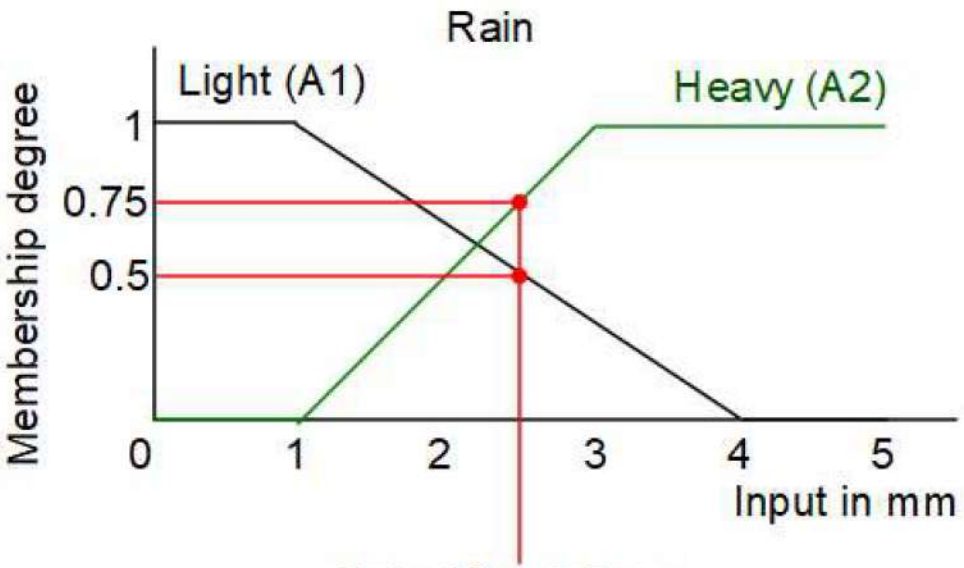
Input reading from sensor:

Rain (X) = 2.5 mm
Road (Y) = 3.5 m3
Speed (Z) = ?

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Case Study – Autonomous Car Speed

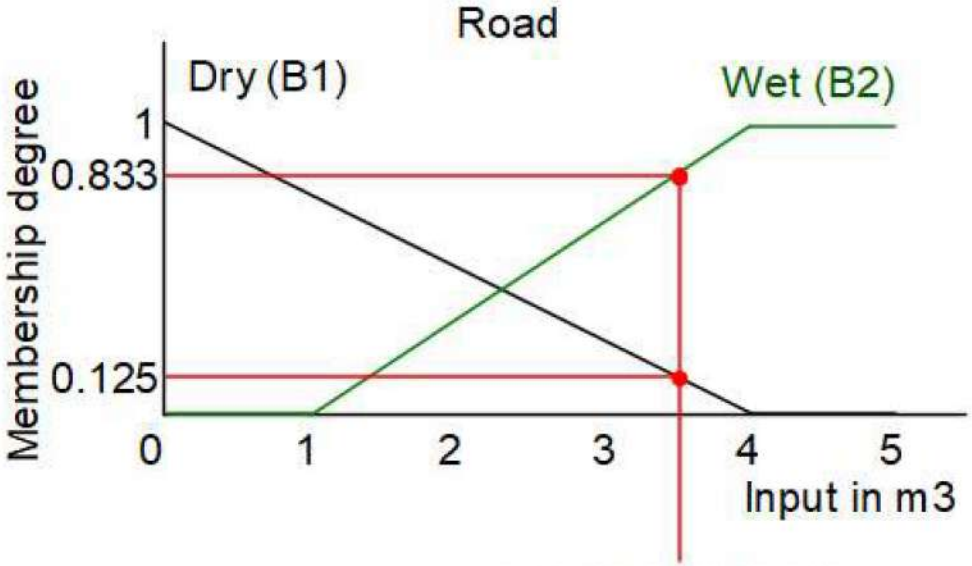
Step 1 - Fuzzification



Rain (X) = 2.5 mm

A1 = 0.5

A2 = 0.75



Road (Y) = 3.5 m3

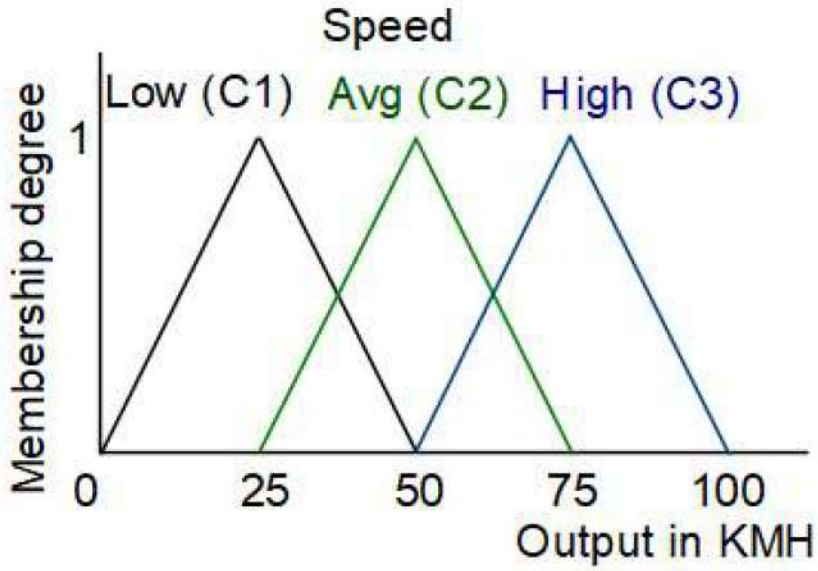
B1 = 0.125

B2 = 0.833

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Case Study – Autonomous Car Speed

Step 1 - Fuzzification

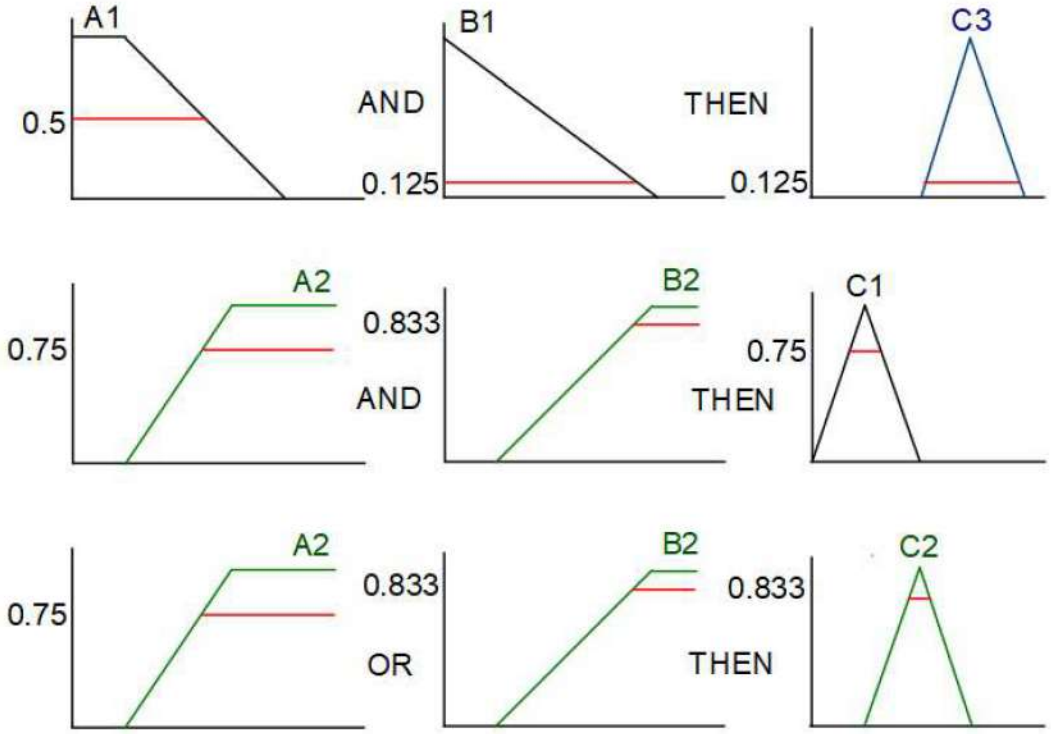


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Case Study – Autonomous Car Speed

Step 2 – Rule Evaluation

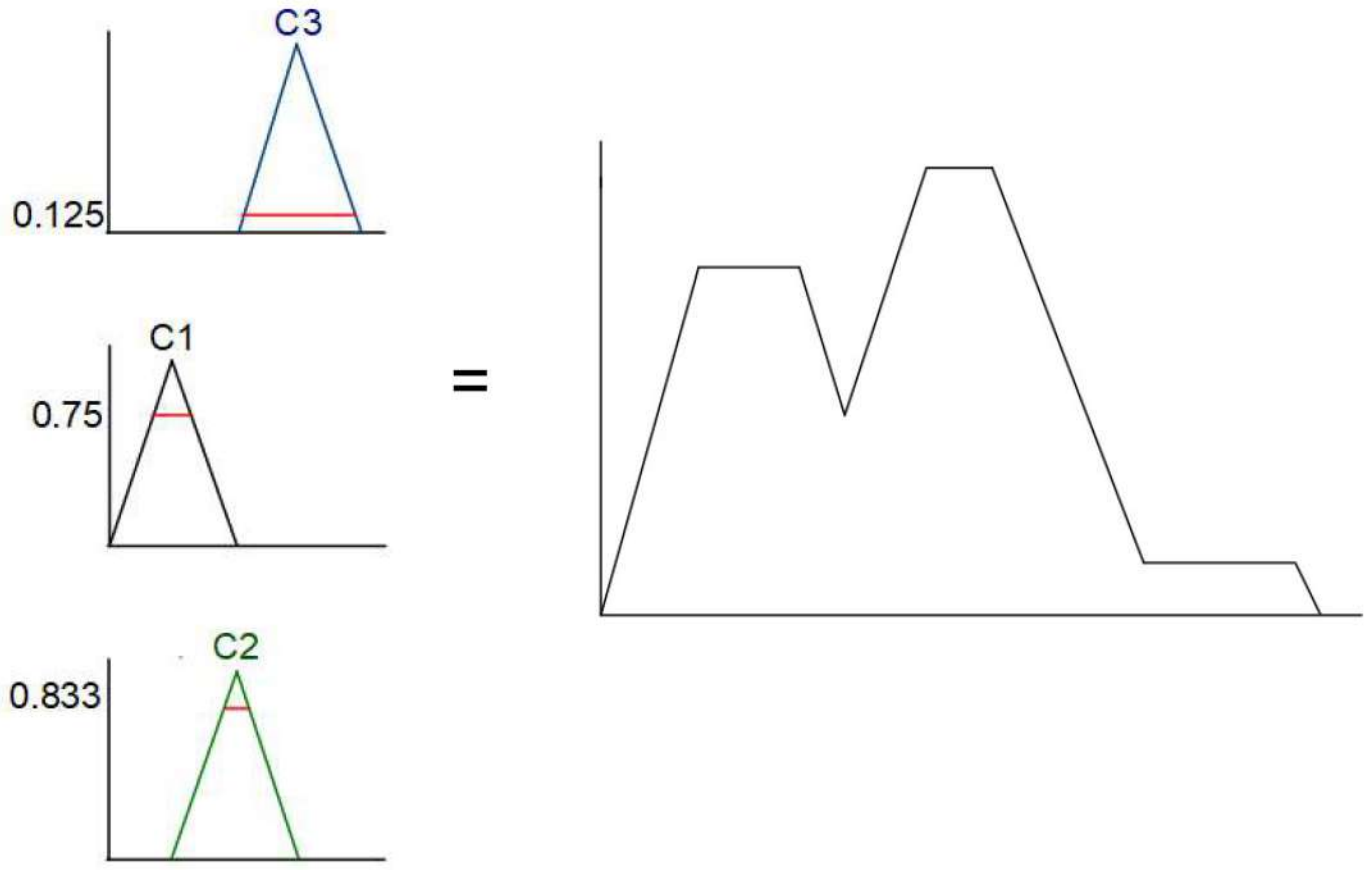
IF Rain is Light (A1 = 0.5) AND Road is Dry (B1 = 0.125) THEN Speed is High (C3 = 0.125)
IF Rain is Heavy (A2 = 0.75) AND Road is Wet (B2 = 0.833) THEN Speed is Low (C1 = 0.75)
IF Rain is Heavy (A2 = 0.75) OR Road is Wet (B2 = 0.833) THEN Speed is Avg (C2 = 0.833)



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Case Study – Autonomous Car Speed

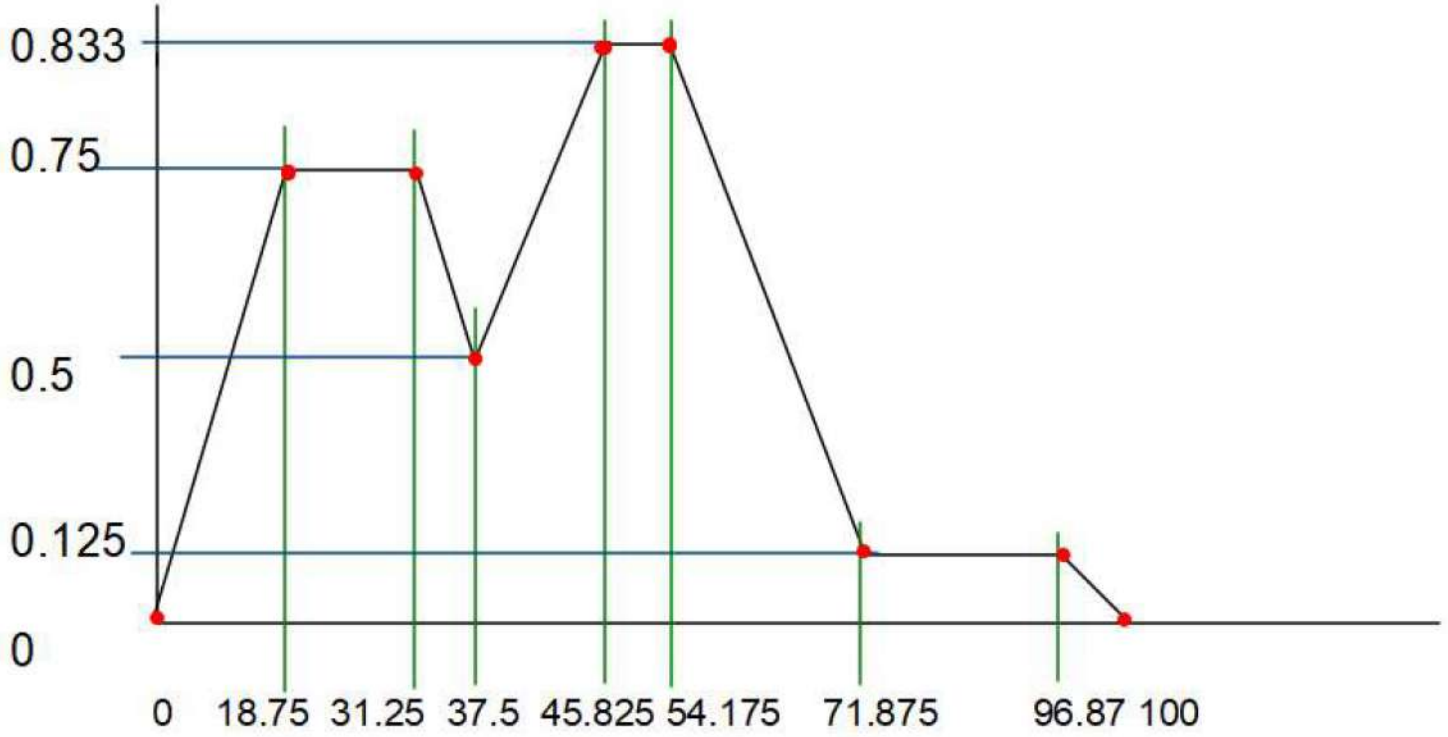
Step 3 – Aggregation



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Case Study – Autonomous Car Speed

Step 4 – Defuzzification

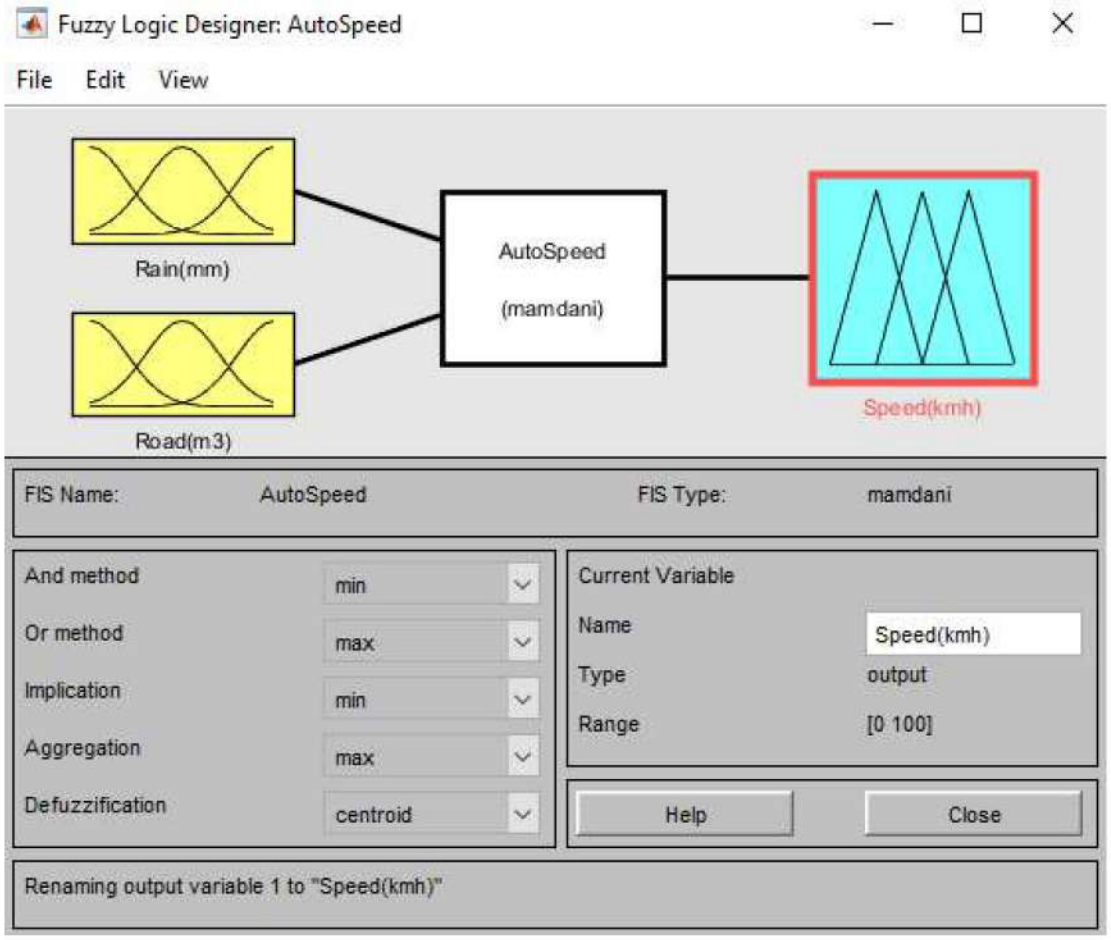


$$\text{COG (Clipping / Mamdani)} = \frac{(18.75*0.75)+(31.25*0.75)+(37.5*0.5)+(45.825*0.833)+(54.175*0.833)+(71.875*0.125)+(96.87*0.125)}{0.75+0.75+0.5+0.833+0.833+0.125+0.125} = 41.016 \text{ kmh}$$

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MATLAB – AutoSpeed.fis

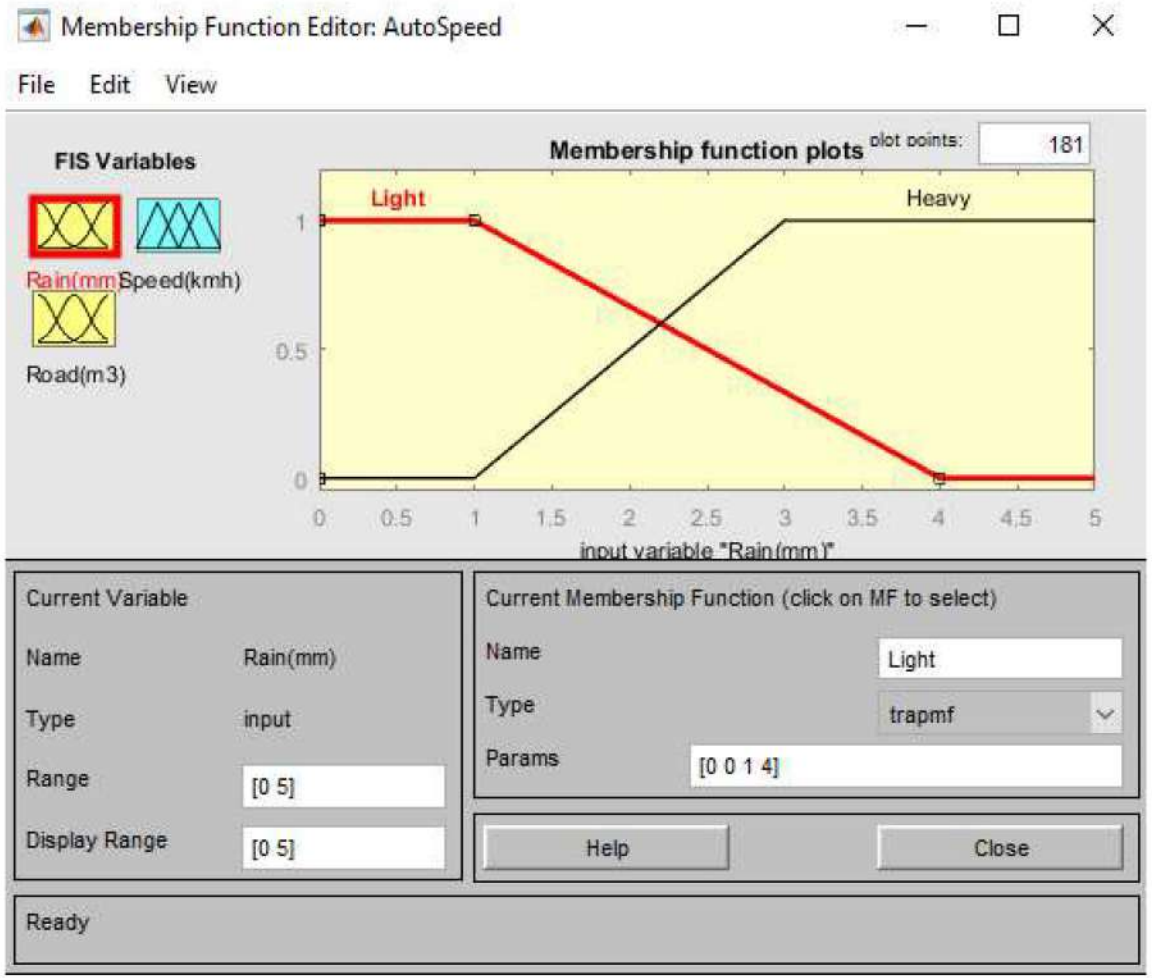
1. Run Fuzzy FIS
2. Create input and output



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MATLAB – AutoSpeed.fis

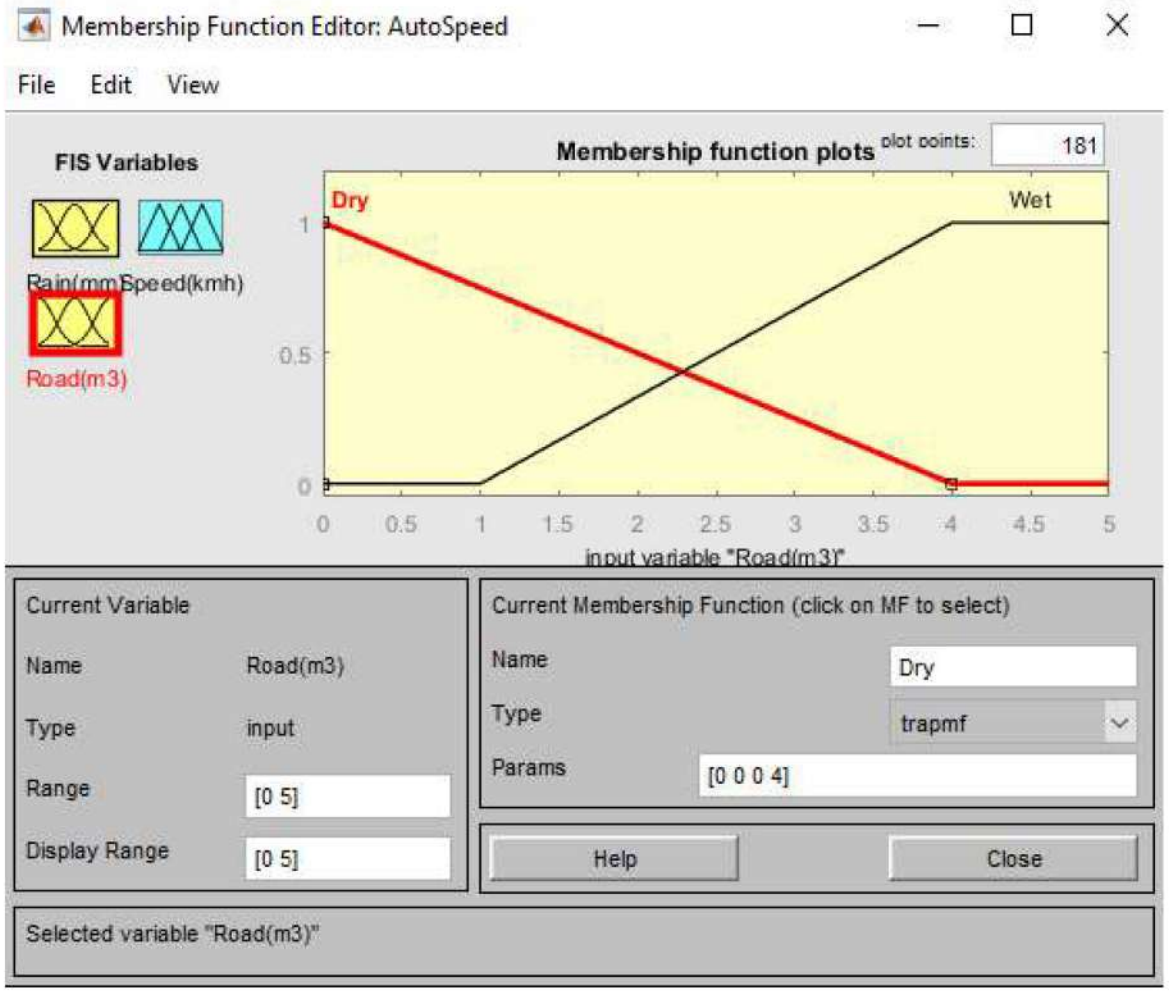
3. Add Membership Function (MF) for Rain - Light & Heavy



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MATLAB – AutoSpeed.fis

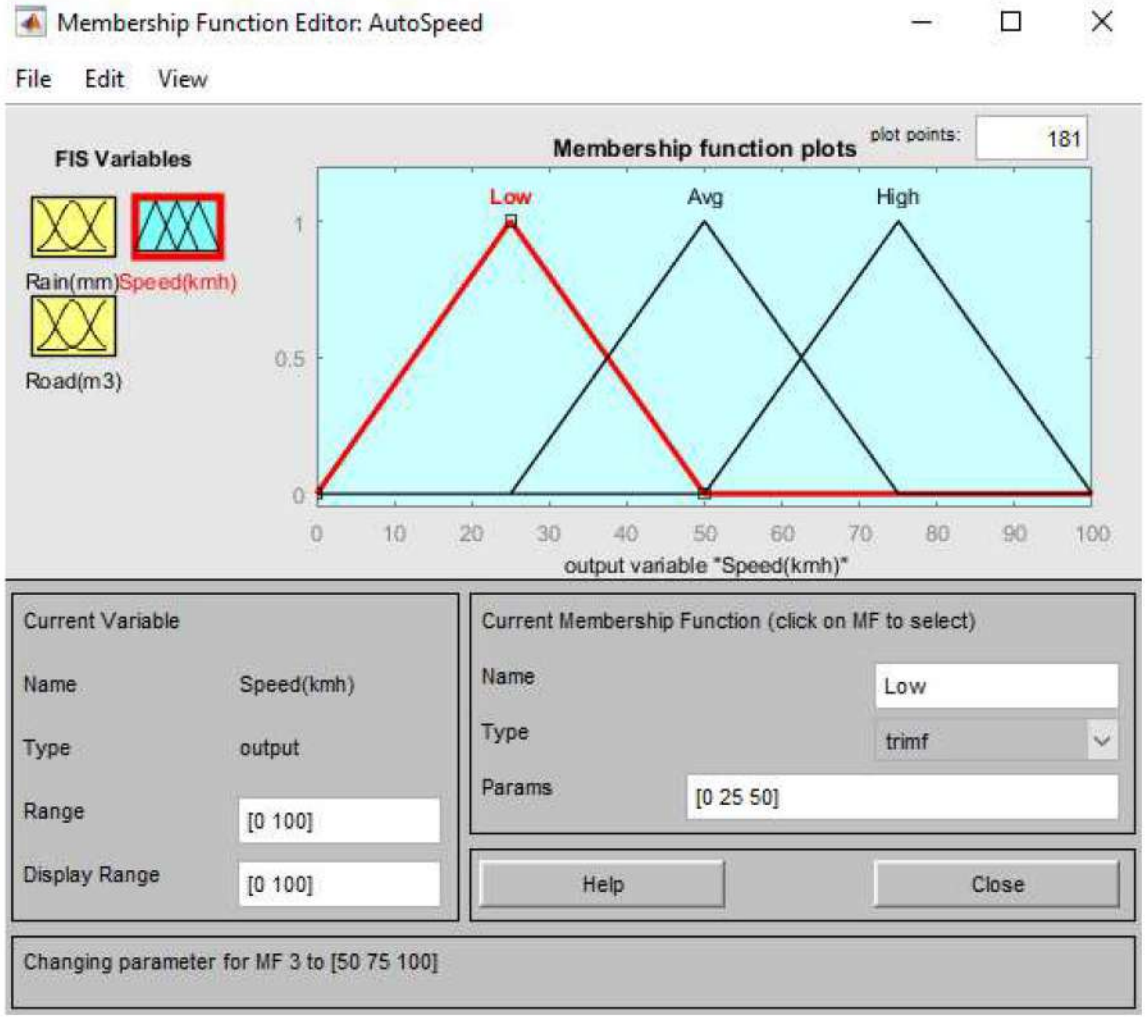
4. Add Membership Function (MF) for Road - Dry & Wet



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MATLAB – AutoSpeed.fis

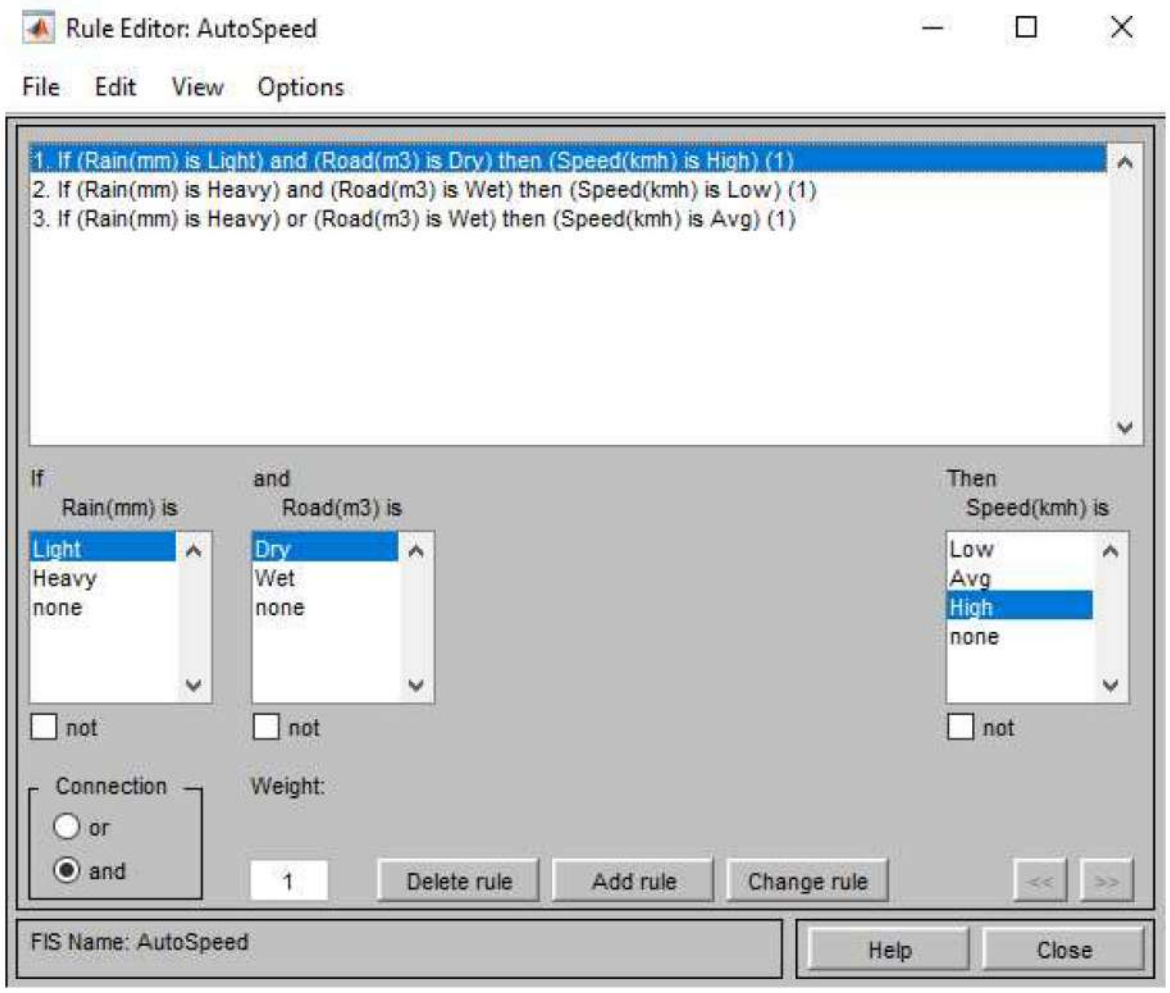
5. Add Membership Function (MF) for Speed – Low, Avg & High



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MATLAB – AutoSpeed.fis

6. Add Rules



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7. Run FIS (Rule Viewer)



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Q & A